

Kyu Hyun Lee

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Academic Appointments

Postdoctoral Scholar, University of California, San Francisco

2020–present

Advisor: Loren M. Frank

Education

California Institute of Technology

2013–2020

Ph.D., Biochemistry and Molecular Biophysics option

Advisor: Markus Meister

Pomona College

2007–2011

B.A., Chemistry (minor: Mathematics), *magna cum laude*

Research Funding and Fellowships

NIH/NEI Pathway to Independence Award (K99/R00)

2025–present

Principal Investigator • K99EY036953

Helen Hay Whitney Foundation Postdoctoral Fellowship

2022–2025

Tianqiao and Chrissy Chen Institute Graduate Innovator Grant

2018

Caltech-Amgen Graduate Fellowship

2014–2015

Publications and Scholarly Products

Preprints

Lee, K., Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M. (2026). A cognitive representation in primary visual cortex modulated by vision. *bioRxiv*. doi: 10.64898/2026.08.29.748029.

Book

Meister, M., **Lee, K.**, and Portuguese, R. (2025). *Mathematics in Biology*. MIT Press.

Peer-reviewed publications

- Guidera, J.A., Gramling, D.P., Comrie, A.E., Joshi, A., Tseng, S.Y., Denovellis, E.L., Smyth, C.N., **Lee, K.**, Zhou, J., Thompson, P., Hernandez, J., Yorita, A., Haque, R., Kirst, C., and Frank, L.M. (2026). Regional specialization in prefrontal cortex manifests in the reliability of task progression codes. *Neuron* 114(6), 1152–1161.e8.
- Lee, K.**^{*}, Denovellis, E.^{*}, Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., Brozdowski, C., Bray, S., Monroe, E., Bak, J., Coulter, M., Sun, X., Broyles, E., Shin, D., Chiang, S., Holobetz, C., Tritt, A., Rübél, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., and Frank, L.M. (2025). Spyglass: a framework for reproducible and shareable neuroscience research. *eLife* 14, RP108089.
- Lee, K.**^{*}, Ni, Y.^{*}, Colonell, J., Karsh, B., Putzeys, J., Pachitariu, M., Harris, T.D., and Meister, M. (2021). Electrode pooling can boost the yield of extracellular recordings with switchable silicon probes. *Nature Communications* 12, 5245.
- Lee, K.**, Tran, A., Turan, Z., and Meister, M. (2020). The sifting of visual information in the superior colliculus. *eLife* 9, e50678.

5. **Lee, K.**, Humbarger, S., Bahnvaldia, R., Sazinsky, M.H., and Crane, E.J., III (2014). Characterization of the mechanism of the NADH-dependent polysulfide reductase (Npsr) from *Shewanella loihica* PV-4: Formation of a productive NADH-enzyme complex and its role in the general mechanism of NADH and FAD-dependent enzymes. *Biochimica et Biophysica Acta - Proteins and Proteomics* 1844(9), 1708–1717.
6. Warner, M.D., Lukose, V., **Lee, K.**, Lopez, K., Sazinsky, M.H., and Crane, E.J., III (2011). Characterization of an NADH-dependent persulfide reductase from *Shewanella loihica* PV-4: Implications for the mechanism of sulfur respiration via FAD-dependent enzymes. *Biochemistry* 50(2), 194–206.

Peer-reviewed conference proceeding

Hermiz, J., Joseph, E., **Lee, K.**, Baldacci, I.A., Chung, J.E., Frank, L.M., Bouchard, K.E., and Denes, P. (2021). The impact of reducing signal acquisition specifications on neuronal spike sorting. *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 5914–5918.

Patent

Meister, M., **Lee, K.**, and Ni, Y. (2024). Methods and systems for electrode pooling. U.S. Patent No. 12,097,033.

* Equal contribution.

Invited Talks and Oral Presentations

Navigating Educational Trajectories in Neuroscience (NIH R25 Program) University of Oregon	2024, 2025
Dynamic Brain Circuits Cluster Databinge University of British Columbia, Djavad Mowafaghian Centre for Brain Health	2024
“Reproducible Workflows for Collaborative Neuroscience” symposium BRAIN Initiative Meeting, NIH	2023
Spike Sorting Workshop Janelia Research Campus	2018
Junior Scientist Workshop on Neural Circuits and Behavior Janelia Research Campus	2017

Selected Poster Presentations

“Visual cues shape an internal task-state representation in primary visual cortex” Lee, K. , Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M. Neurobiology of Cognition Gordon Research Conference, Waterville Valley, New Hampshire	2026
“A high-resolution spatial map in the primary visual cortex of freely behaving rats” Lee, K. , Adenekan, P., Gao, F., Yorita, A., Haque, R., Eden, U., Scanziani, M., and Frank, L.M. Neuroscience 2025, Society for Neuroscience, San Diego, California	2025
“Cognitive representations in primary visual cortex” Lee, K. , Adenekan, P., Gao, F., Yorita, A., Haque, R., Eden, U., Scanziani, M., and Frank, L.M. Neural Coding and Dynamics, Lake Conferences, Seattle, Washington	2025
“Spyglass: a data analysis framework for reproducible and shareable neuroscience research” Lee, K. , Denovellis, E., Ly, R., Magland, J., Soules, J., Comrie, A., Guidera, J., Nevers, R., Gramling, D., Adenekan, P., Coulter, M., Bak, J., Monroe, E., Tritt, A., Rübel, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., Aery Jones, E., Giocomo, L., and Frank, L. Neuroscience 2022, Society for Neuroscience, San Diego, California	2022
“Visual sifting in the superior colliculus” Lee, K. , Tran, A., Turan, Z., and Meister, M. Simons Collaboration on the Global Brain, New York City, New York	2019

Workshops Attended

NWB Developer Days Janelia Research Campus	2023
NeuroData ReHack Allen Institute for Brain Science	2022
Spike Sorting Hackathon Center for Computational Biology, Flatiron Institute	2022
Mountainsort User Workshop Center for Computational Biology, Flatiron Institute	2018

Honors and Awards

Phi Beta Kappa	2011
Robert Rowan III Memorial Award	2011

Research Mentoring

Denisse Morales-Rodriguez , Graduate Student, UCSF Optogenetic manipulation of mPFC during sleep.	2024–present
Scott Gao , Research Associate, UCSF Hippocampus–visual cortex interactions.	2024–present
Philip Adenekan , Research Associate, UCSF Hippocampus–visual cortex interactions.	2022–2024
Yangsun Hwang , Rotation Student, UCSF Chronic tracking of neurons recorded with polymer probes.	2023
Roshanak Gonzalez , Rotation Student, UCSF Testing an optoelectronic neural probe.	2022
Daryl Lawrence , Rotation Student, UCSF Denoising signals from a neural-recording amplifier chip.	2021

Teaching Experience

Teaching Assistant Division of Biology and Biological Engineering, California Institute of Technology	
Principles of Neuroscience (Fall term)	2018
Tools of Neurobiology (Fall term)	2016
Mathematics in Biology (Fall term)	2015
Brain Circuits (Winter term)	2014